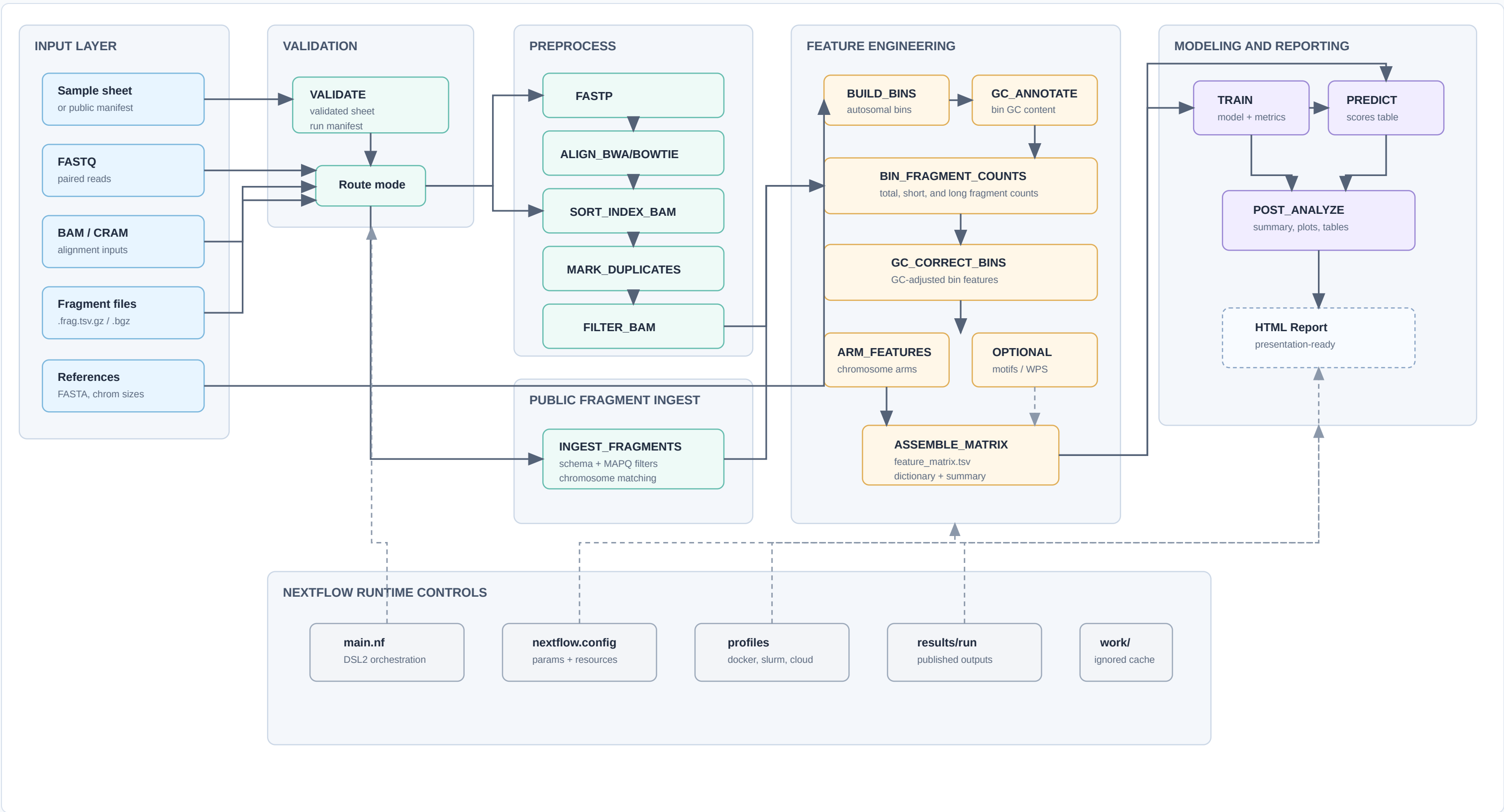


Fragmentomics NF Architecture Workflow

End-to-end view of input routing, preprocessing, public fragment ingest, feature engineering, baseline modeling, prediction analysis, and Nextflow runtime controls.

Research pipeline architecture | source: main.nf + modules/local + bin



Primary path

Sample metadata is validated first, then the pipeline routes sequencing inputs through preprocessing or public fragment files through direct ingest. Both paths converge before bin counting and matrix assembly.

Tracked outputs

Small representative reports are kept in the repository. Heavy outputs such as raw fragments, work cache, feature matrices, reference genomes, and model binaries remain ignored.

Runtime model

Nextflow orchestrates module wrappers, Python scripts implement domain logic, and profiles configure local, container, cluster, or cloud execution.